Week 4 Online class

Worksheet 2 (Q2, 12, 13, 14)

Question 2. Give an example or prove the following is impossible.

(a) A sequence with an infinite number of zeros that does not converge to zero.

(b) A sequence with an infinite number of zeros that converges to a non-zero limit.

(c) A divergent sequence such that for all $n \in \mathbb{N}$, there are n consecutive zeros somewhere in the sequence.

Answer. (a) $\left(1 + (-1)^n\right)$. (c) $(1, 0, 1, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 1, \ldots)$.

Question 2. Show that this statement is false.

(b) A sequence with an infinite number of zeros that converges to a non-zero limit.

Answer. For example, sequence like these:

$(0,1,0,1,0,1,0,1,\ldots)$, $(1,0,1,0,0,1,0,0,0,1,0,0,0,0,1,\ldots)$.

Proof. Let (a_n) be a convergent sequence with $\lim\left(a_n\right) = a \neq 0$.

Then $\forall \varepsilon > 0\ \exists N \in \mathbb{N}$ such that $n > N \Rightarrow \left|a_n - a\right| < \varepsilon\ \forall n \in \mathbb{N}$.

Choose $\varepsilon = \frac{|a|}{2} > 0\ (\because a \neq 0)$.

Now $\exists M \in \mathbb{N}$, $M > N$ such that $a_M = 0$

(because there are infinite number of zeros).

So, $\left|a_M - a\right| = \left|-a\right| = \left|a\right| < \varepsilon = \frac{|a|}{2}$, which is a contradiction.

Question 12.

Let (a_n) be a sequence with $a_n \in \mathbb{Z}\ \forall n \in \mathbb{N}$.

(a) Is it possible for $\lim\left(a_n\right) = 3.5$?

Answer. The answer is NO.

Proof. Suppose $\lim\left(a_n\right) = 3.5$. Then

$$\forall \varepsilon > 0 \ \exists N \in \mathbb{N} \ni n > N \Rightarrow \left|a_n - 3.5\right| < \varepsilon \ \forall n \in \mathbb{N}.$$

Choose $\varepsilon = \frac{1}{4}$ we then have

$$3.25 = 3.5 - 0.25 < a_n < 3.5 + 0.25 = 3.75.$$

In other words, $a_n \notin \mathbb{Z}$, a contradiction.

(b) If $a_n \neq a_m$ for all $n \neq m$, prove that $\left(a_n\right)$ does not converge.

Proof. Suppose $a_n \neq a_m$ for all $n \neq m$ but $\left(a_n\right)$ converges to a.

Then $\exists M \in \mathbb{N}$ such that whenever $n, m > M$

$$\left|a_n - a\right| < \tfrac{1}{2} \text{ and } \left|a_m - a\right| < \tfrac{1}{2}.$$

Thus, implies that

$$0 \underset{\because a_n \neq a_m}{<} \left|a_n - a_m\right| = \left|\left(a_n - a\right) - \left(a_m - a\right)\right| \leq \left|a_n - a\right| + \left|a_m - a\right| < 1.$$

Then $\left|a_n - a_m\right| \notin \mathbb{N}$, a contradiction.

Remark: Note that the hypothesis of the statement

can be relaxed to $a_n \neq a_m$ for all $n \neq m$ when $n, m > N$

for some $N \in \mathbb{N}$.

(c) If (a_n) converges, what can be said about this sequence?

Answer. The sequence $\left(a_n\right) = \left(a_1, \ldots, a_N, c, c, c, \ldots\right)$,

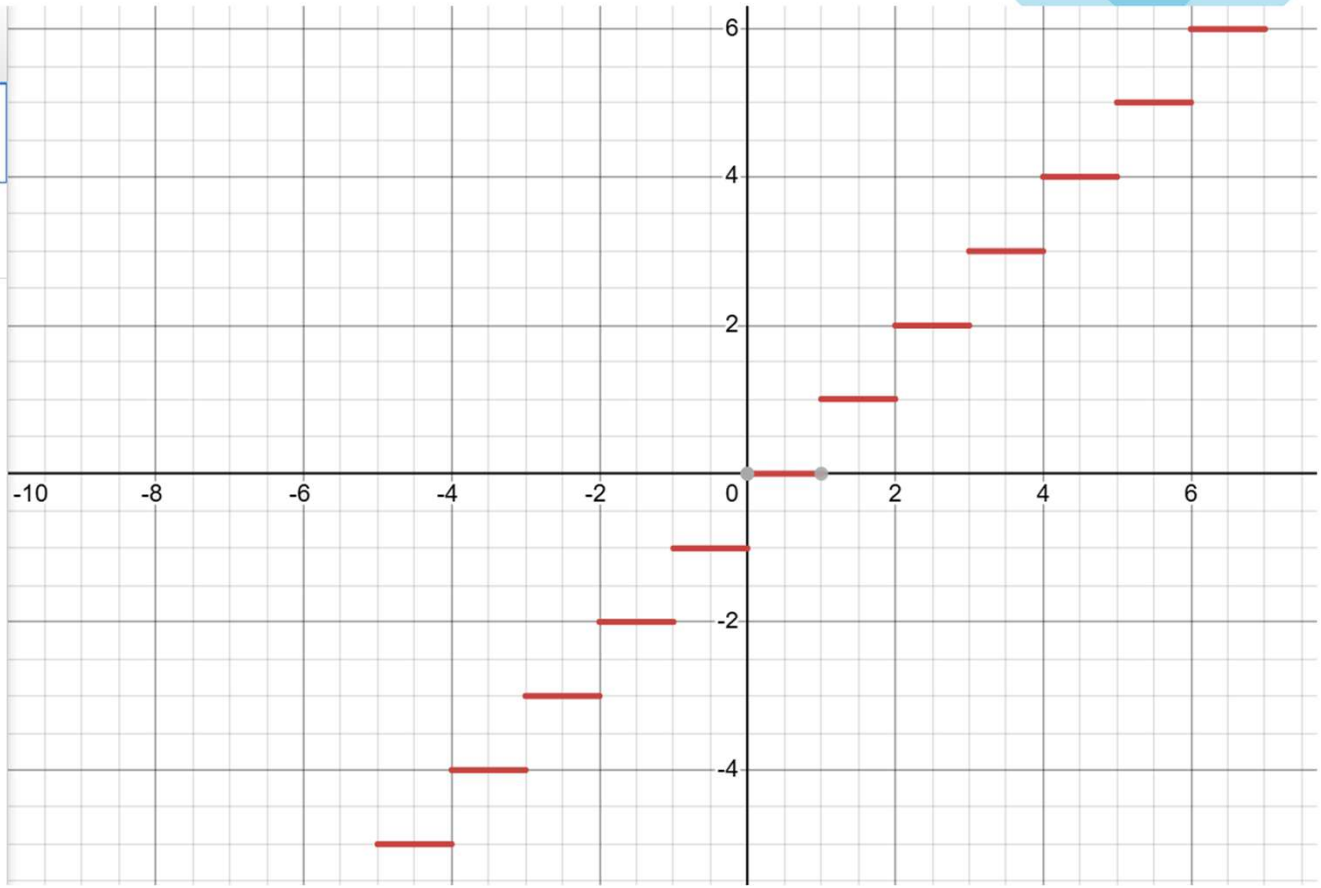
i.e., $a_1, \ldots, a_N$ are arbitrary and $a_n = c$ (a constant integer) for all $n > N$.

Question 13. Let $\lfloor x \rfloor$ be the greatest integer $\leq x$ (floor function).

For example, $\lfloor \pi \rfloor = 3$ and $\lfloor a \rfloor = a$ if $a \in \mathbb{Z}$.

For each sequence, find $\lim\left(a_n\right)$ and prove it by definition.

$y = \text{floor}(x)$



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(a) $a_n = \left\lfloor \frac{5}{n} \right\rfloor$, $\lim\left(a_n\right) = 0$

Proof. $a_n = \left\lfloor \frac{5}{n} \right\rfloor = 0$ for all $n > 5$. Let $\varepsilon > 0$. Then

$$n > 5 \Rightarrow \left| \left\lfloor \frac{5}{n} \right\rfloor - 0 \right| = 0 < \varepsilon \quad \forall n \in \mathbb{N} \setminus \{1,2,3,4,5\}.$$

(b) $a_n = \left\lfloor \frac{12+4n}{3n} \right\rfloor$, $\lim(a_n) = 1$

Proof. $a_n = \left\lfloor \frac{4}{n} + \frac{4}{3} \right\rfloor = 1$ for all $n > 6$. Let $\varepsilon > 0$. Then

$n > 6 \Rightarrow \left| \left\lfloor \frac{4}{n} + \frac{4}{3} \right\rfloor - 1 \right| = |1 - 1| = 0 < \varepsilon \quad \forall n \in \mathbb{N} \setminus \{1,2,3,4,5,6\}$.

Definition of a function

Let A and B be nonempty sets.

A *function* f from A to B,

written as $f : A \to B$,

is a mapping of each element in A

to a unique element in B.

Definition of an injective function 单射

A function f is said to be *one-to-one*, or *injective*, if

$$f(a) = f(b) \Rightarrow a = b$$

for all a, b in the domain of f.

Equivalently, $a \neq b \Rightarrow f(a) \neq f(b)$.

Definition of a surjective function 满射

The function $f : A \to B$ is called *onto*, or *surjective*, if for every element b in B there is an element a in A such that $f(a) = b$.

Definition of a bijective function 双射

The function f is a *one-to-one correspondence*,
or *bijective*, if it is both one-to-one (injective) and onto (surjective).
When $f : \mathbb{N} \to \mathbb{N}$ is bijective, we call f a permutation of $\mathbb{N}$.

Question 14. Let $f : \mathbb{N} \to \mathbb{N}$ be a bijection (f is called a permutation of $\mathbb{N}$). Show that if $\left(a_n\right)$ converges to $a \in \mathbb{R}$, then $\left(a_{f(n)}\right)$ also converges to a.

(This shows that any reordering of a convergent sequence still converges to the same point.)

Proof. Theorem 9.3: $\left(a_n\right)$ converges if and only if $\left(a_n\right)$ is Cauchy.

Then for all $\varepsilon > 0$, there exists a $N \in \mathbb{N}$ such that

$$\left|a_n - a_{f(n)}\right| < \tfrac{\varepsilon}{2} \text{ and } \left|a_n - a\right| < \tfrac{\varepsilon}{2} \; \forall n, f(n) > N .$$

It follows that

$$\left|a_{f(n)} - a\right| = \left|a_{f(n)} - a_n - (a - a_n)\right| \leq \left|a_{f(n)} - a_n\right| + \left|a_n - a\right| < \varepsilon ,$$

whenever $f(n) > N$.